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NPTEL Course on
"Computer System Optimization for ML"

Lecture #38

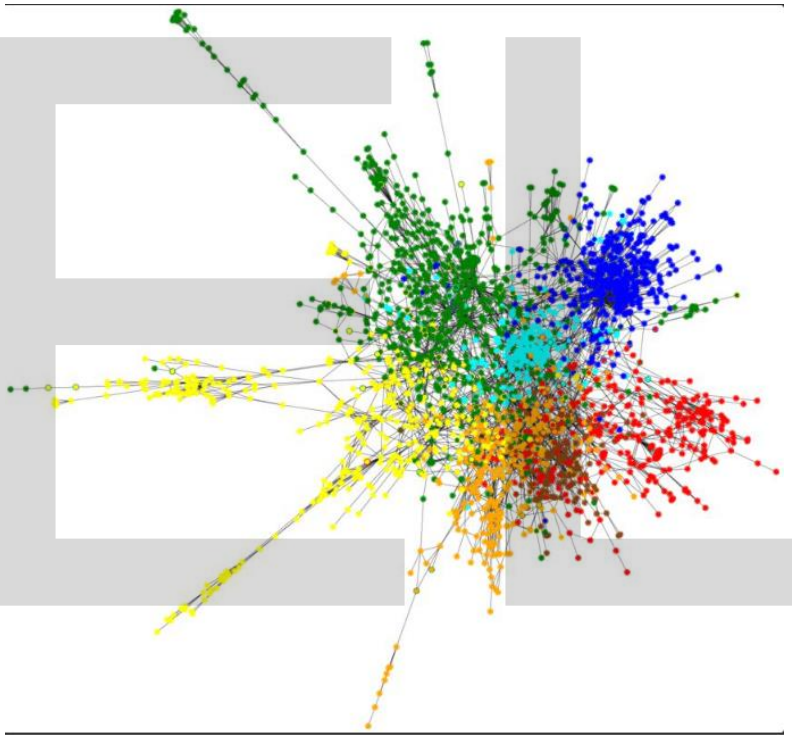
Agenda

- **Graph Neural Network**

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Representation of Cora Dataset

- 2708 scientific publications
- Classified into one of seven classes
  - Each colour denotes a class
- The network consists of 5429 links (citations)
- Each publication is represented by a 0/1-valued word vector indicating absence/presence of the corresponding word from the dictionary (features)
- The dictionary consists of 1433 unique words





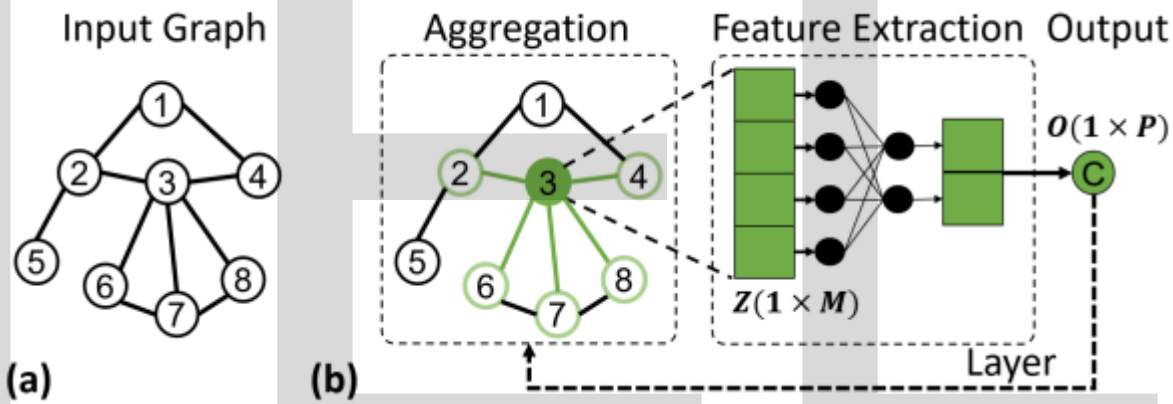
GNN Training and Inference

- **Chalk and Talk**

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Classification of Graph Nodes through GNN

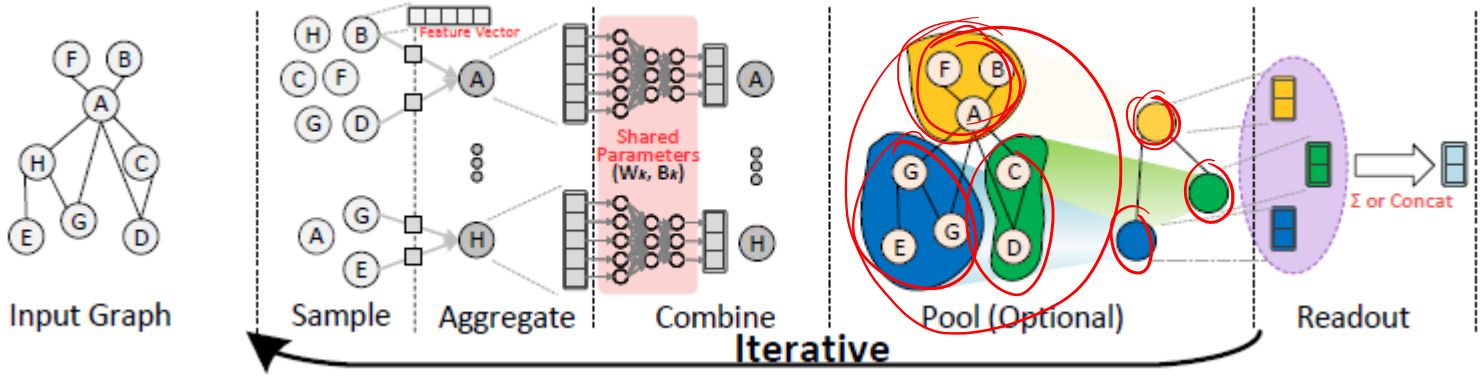
- **Two phases**
  - Aggregation
  - Combination/Feature Extraction
- **Aggregate function aggregates multiple feature vectors from neighbours to form one single feature vector**
  - Different GNNs use different function
- **Combination function transforms the feature vector of each vertex to another feature vector using a MLP**
- **MLP parameters are shared between vertices**



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Classification of Graph Nodes through GNN

- **After several iterations, the features are used for final prediction od classification**
- **To reduce computational complexity, the Sample function is usually applied before Aggregate**
  - Sample a subset from the neighbour vertices of each vertex as new neighbour
- **Sometimes 'Pool' follows the combine function to transform the original graph to a smaller graph**
- **For node classification, vertex features vectors at the last iteration are used for prediction**
- **For graph classification, a ReadOut function further aggregates the vectors at the last iteration to obtain the entire graph's representation vector**



Different Aggregation and Combination Function

- **Notations**
  - a_v^k: aggregation at node-v at the iteration k
  - D_v: degree of node-v
  - h_v: feature vector at node-v
- **Graph Convolutional Network** (GCN)
  - $a_v^k = \left(\sum \frac{1}{(D_v D_u)^{\frac{1}{2}}} h_u^{k-1}\right)$, $h_v^k = ReLU(W^k a_v^k + b^k)$
- **GraphSage: uniform neighbour sampling**
  - $a_v^k = Mean(\{h_v^{k-1}\} \cup \{h_u^{k-1}, \forall u \in S(v)\})$, $h_v^k = ReLU(W^k a_v^k + b^k)$
- **GINConv**
  - $a_v^k = (1 + \epsilon_k) h_v^{k-1} + \sum h_u^{k-1}$, $h_v^k = MLP^k(W^k,\ a_v^k,\ b^k)$

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Summary of GNN Computations

- Several operations
  - Aggregation, Combination, Pooling, Readout
- Except combination, all the operations are graph structure-dependent, which involve graph processing
- Combination is a typical MLP
- Sampling is to sample a subset from neighbours
- Aggregation aggregates the features from one hop neighbours
- Pooling acts like a pooling in CNN
- Readout can be a simple summation across vertices; can be seen as extreme aggregation
- Aggregation and Combination are the two major phases

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THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)